

ETHAN KIM

CONTACT INFORMATION

336-709-9546
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Chapel Hill, NC

EDUCATION

University of North Carolina at Chapel Hill
Bachelor of Science Computer Science
& Minor in Statistics and Operations
Expected 2029

Associates in Arts from Guilford Technical
Community College

Relevant Courses

Fall 2026:

COMP 210 - Analysis of Data Structures

COMP 283 - Discrete Structures

STOR 305 - Intro Decision Analytics

MATH 233 - Multi Vari Calc 1 (Calc 3)

APPL 101 - Intro to Engineering

Spring 2027:

COMP 211 - Systems Fundamentals

COMP 301 - Foundations of Programming

Comp 455 - Models of Languages and
Computation

MATH 347 - Linear Algebra for Applications

STOR 235 - Introduction to Probability

SKILLS

Languages

- Java
- Python
- JavaScript
- TypeScript
- HTML
- CSS
- Lua

Frameworks

- React
- Next.js
- FastAPI
- Flask
- Websockets
- SocketIO
- OpenCV
- YOLO

Robotics

- Kalman Filters for
Sensor Fusion
- Control Systems
- Mapping
- Motion Planning
- Inverse Kinematics
- Computer Vision

Hardware

- Jetson Orin Nano &
other Linux devices
- Arduino
- Realsense Depth
Cameras
- GPS & RTK GPS
- Radio
- Lidar

CERTIFICATIONS

Java Programming Certification from
Guilford Technical Community College

LinkedIn



ethan-kim-590275394

GitHub



DoctorKrill08

Portfolio



https://ethankimufgonline.com

EXPERIENCE AND PROJECTS

Intern at North Carolina Agriculture and Technical State University

Helped PHD students create and implement **software** for
Outdoor Agricultural mobile robots with **sensor fusion** for
localization, mapping, obstacle avoidance and a **web based**
dashboard with telemetry, mapping, camera perception and
robot control.

- Developed **autonomous navigation** software for mobile
robots using **LiDAR, depth cameras, and RTK GPS**.
- Designed **Kalman filter** sensor fusion integrating RTK GPS,
IMU, and **wheeled odometry** for **outdoor localization**.
- **Combined LiDar and Depth Cameras perception** with
localization to create a **Dynamic Obstacle Map**.
- Implemented **Dynamic Window Approach** local path
planning for **obstacle avoidance** and navigation.
- Built a **Web interface** for robot telemetry and control with a
python fastAPI backend, uvicorn server, a React TSX
frontend, and with Websockets as the communication
protocol.
- Trained and integrated **YOLO object detection** models for
robotic arm use.
- Developed **inverse kinematics algorithms** for robotic arm
positioning and movement.

Undertale Fighting Game

- Created, designed and **programmed** a multiplayer Roblox
game with over **6 million visits** and over **2 million unique**
players
- Designed gameplay, visual effects, movement, and
interfaces.
- Created and managed an online community supporting
hundreds of players

First Tech Challenge Team 10195 Captain and Lead Programmer

- Led software development for a competitive First Tech
Challenge Robot.
- Implemented **PID Control** for Motor Control based on
Encoder and Camera data.
- Created a **Kalman Filters** for Sensor fusion between
Wheeled Odometry, IMU, and April Tags.
- As lead programmer created robust autonomous routines
and achieved North Carolina **Autonomous** Offensive Power
Rating **State Record** in 2024 - 2025.
- As captain led the team to **win their first competition** since
2017 in the 2025 - 2026 season.